

HEARNE, TX, USA

WMO: 720303

Lat: 30.872N

Lon: 96.622W

Elev: 285

StdP: 14.55

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53973

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.4	29.1	12.7	10.6	38.9	17.6	13.5	40.0	25.4	50.8	23.2	50.0	5.2	330	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.0	99.8	74.2	98.2	74.1	96.2	74.4	80.5	89.5	79.3	88.3	78.2	87.4	7.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	150.4	83.6	77.2	143.2	82.2	76.2	138.0	81.7	44.3	90.0	42.9	88.3	41.8	87.1	97.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	18.2	16.2	DB	20.8	105.5	4.3	5.7	17.7	109.6	15.1	113.0	12.7	116.2	9.5	120.4	
			WB	18.3	82.6	4.2	6.2	15.3	87.0	12.9	90.7	10.5	94.2	7.5	98.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.3	50.0	54.4	61.1	67.4	74.9	81.8	84.1	84.5	79.4	69.3	59.7	51.6	(6)
(7)		DBStd	14.50	9.97	10.40	9.53	7.49	6.01	3.43	3.65	3.77	5.40	8.08	9.86	10.20	(7)
(8)		HDD50	345	123	66	21	2	0	0	0	0	1	27	105		(8)
(9)		HDD65	1749	477	318	187	60	7	0	0	0	1	52	213	434	(9)
(10)		CDD50	7008	122	189	367	523	771	955	1057	1070	881	600	319	154	(10)
(11)		CDD65	2935	10	21	67	131	314	505	592	605	432	185	55	18	(11)
(12)		CDH74	31498	67	125	438	1090	2900	5638	7043	7396	4554	1759	398	90	(12)
(13)		CDH80	15256	5	13	78	265	1120	2807	3824	4151	2272	626	83	12	(13)
(14)	Wind	WSAvg	6.0	6.4	7.6	7.7	7.6	6.7	5.9	5.2	4.8	4.2	5.0	5.7	5.9	(14)
(15)	Precipitation	PrecAvg	38.3	3.1	3.0	3.1	3.2	3.7	3.5	2.1	2.4	3.3	4.3	3.3	3.4	(15)
(16)		PrecMax	56.5	10.6	8.3	7.7	9.6	8.0	12.7	8.0	5.7	8.1	11.5	9.5	11.2	(16)
(17)		PrecMin	23.2	0.2	0.2	1.1	0.9	0.0	0.0	0.0	0.1	0.6	0.0	0.2	0.7	(17)
(18)		PrecStd	9.0	2.2	2.1	1.5	2.0	2.2	2.8	1.8	1.9	2.1	3.3	2.5	2.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.9	81.7	87.5	90.1	95.0	99.8	108.5	103.8	99.9	92.9	86.2	81.4	(19)
(20)		MCWB	62.0	64.2	71.1	69.7	72.4	72.2	84.1	74.4	71.2	71.7	72.6	63.3		(20)
(21)		2%	DB	75.2	78.2	81.8	86.2	91.1	97.0	99.5	100.0	97.0	89.7	82.0	76.6	(21)
(22)		MCWB	61.3	63.9	67.2	69.3	72.8	72.4	75.0	74.7	72.9	71.9	69.1	64.7		(22)
(23)		5%	DB	71.7	73.7	79.0	82.5	89.3	94.7	97.3	98.5	94.1	86.4	78.9	72.5	(23)
(24)		MCWB	61.3	63.5	65.9	69.0	73.1	73.8	74.5	74.9	73.5	70.7	66.3	63.5		(24)
(25)		10%	DB	67.5	70.9	75.8	80.4	86.6	91.6	95.0	96.4	91.2	83.8	74.9	69.6	(25)
(26)		MCWB	59.3	62.9	64.8	67.6	72.3	75.0	75.4	75.1	72.9	68.7	65.9	63.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.1	70.2	73.9	75.9	79.3	81.6	87.7	81.0	79.9	77.3	79.7	70.9	(27)
(28)		MCDWB	72.4	74.8	80.9	83.1	86.8	90.0	101.6	92.2	87.6	84.3	80.4	74.0		(28)
(29)		2%	WB	66.2	68.3	71.1	73.9	77.3	79.6	81.1	79.7	78.3	75.4	72.5	68.5	(29)
(30)		MCDWB	71.3	73.4	76.9	80.3	84.2	87.5	90.9	90.8	86.6	83.4	77.4	72.6		(30)
(31)		5%	WB	64.1	66.4	69.2	72.3	75.8	78.4	79.3	78.3	77.3	74.1	70.2	66.2	(31)
(32)		MCDWB	69.3	71.3	75.0	79.0	83.3	86.3	88.3	88.9	86.1	81.2	75.0	70.8		(32)
(33)		10%	WB	60.6	64.4	67.4	70.8	74.6	77.3	77.9	77.4	76.3	72.7	68.0	63.7	(33)
(34)		MCDWB	66.1	69.1	73.1	77.7	82.0	85.4	87.7	87.7	85.1	79.4	73.5	68.4		(34)
(35)	Mean Daily Temperature Range	MDBR	22.3	21.3	22.1	22.0	20.0	19.8	21.0	22.0	21.8	24.0	22.8	21.5		(35)
(36)		5% DB	MCDBR	27.1	23.8	22.7	22.9	22.0	23.7	24.7	24.4	25.0	24.3	24.1	24.3	(36)
(37)		MCWBR	15.6	13.7	11.4	10.5	7.7	6.0	5.7	5.1	7.0	9.8	13.0	15.8		(37)
(38)		5% WB	MCDBR	19.9	19.0	18.6	17.9	17.5	17.2	20.7	21.0	19.1	19.7	19.4	20.1	(38)
(39)		MCWBR	15.0	13.4	12.2	10.2	8.1	7.0		5.4	6.2	10.1	13.2	15.8		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.342	0.365	0.412	0.454	0.445	0.461	0.450	0.417	0.374	0.364	0.326	(40)	
(41)		taud	2.504	2.462	2.400	2.285	2.214	2.288	2.275	2.298	2.342	2.444	2.413	2.504	(41)	
(42)		Ebn at Noon	288	292	291	280	267	268	264	266	271	277	270	281	(42)	
(43)		Edh at Noon	28	32	36	43	46	43	43	41	38	32	30	27	(43)	
(44)	All-Sky Solar Radiation	RadAvg	921	1044	1399	1706	1832	2047	2079	1954	1640	1341	994	808	(44)	
(45)		RadStd	91	194	128	129	143	156	150	113	143	182	114	103	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155

Nomenclature: See separate page